
RENTASALUD: A WEB-BASED INTERACTIVE ATLAS OF SOCIAL INEQUALITIES AND LIFE EXPECTANCY IN ANDALUSIA (SOUTHERN SPAIN)

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ABSTRACT

Background Interactive tools linking income and life expectancy at fine geographical scales are scarce for southern Europe. **Methods** We linked INE census-section income data (49,888 section-years, 2015–2022) with BDLPA cohort life tables (~637,000 individuals, follow-up 2011–2023) for Andalusia, Spain. Correlation and clustering assessed the income–longevity association. **Results** Life expectancy was 79.5 (men) and 84.4 years (women). Eliminating tumours would yield the largest gain for men (3.93 years), while heart disease accounted for the largest gain among women (2.89 years). The income–longevity correlation was weak ($r=0.21$, 95%CI 0.59 to 0.81). An R Shiny application integrates all findings in six interactive tabs. **Conclusion** RENTASALUD provides an open tool for exploring health inequalities. Provincial aggregation masks steep census-tract gradients reported nationally. **Keywords:** health inequalities; life expectancy; spatial epidemiology; interactive atlas; Shiny; Andalusia

1 Introduction

It is well known that people with more money tend to live longer, and this pattern holds across most European countries (Marmot 2010; Mackenbach et al. 2015). In Southern Europe — including Spain — the gap between rich and poor is somewhat narrower than in Northern or Eastern Europe. This has been linked to the Mediterranean diet, strong family and community networks, and universal healthcare (Mackenbach et al. 2018; Regidor et al. 2012). But even in Spain, educational disparities still account for roughly one in five deaths — about 80,000 each year — mostly from heart disease, lung cancer, respiratory illnesses, and accidents (Regidor et al. 2014).

How big is the gap when we look at small areas like neighbourhoods or census tracts? A recent national study by Redondo-Sánchez and colleagues (Redondo-Sánchez et al. 2022) found that women and men living in Spain’s most deprived neighbourhoods lived 3.2 and 3.8 years less, respectively, than those in the wealthiest areas. This gap was wider for men and varied by region, with northern Spain and provincial capitals faring better.

Andalusia, home to about 8.5 million people, is Spain’s largest region and one of its most economically disadvantaged. Despite this, there are few interactive tools that let people explore how income, demographics, and life expectancy come together at the neighbourhood level. Most existing maps are either too coarse (showing only whole provinces) or lack the interactive features that would make them useful for policymakers, researchers, or the public.

RENTASALUD was built to fill this gap. It is an open-source web atlas that combines detailed census-section data from Spain’s National Statistics Institute (INE, 2015–2022) with life expectancy estimates from the Longitudinal Database of the Andalusian Population (BDLPA), which follows about 637,000 people from the 2011 census through to 2023. Our aims were: (i) to estimate life expectancy by province and sex; (ii) to see how much longer people would live if certain causes of death were eliminated; (iii) to examine the relationship between income and life expectancy; and (iv) to build an interactive tool that anyone can use.

2 Methods

2.1 Data sources

We used two main datasets. The first comes from the INE’s *Atlas de Distribución de Renta de los Hogares*, covering all eight Andalusian provinces from 2015 to 2022. It includes nearly 50,000 census section-year observations with information on median income, age structure, household composition, and population size — all at the neighbourhood level.

The second dataset is the BDLPA, which follows a random 10% sample of people who lived in Andalusia at the time of the 2011 census (~637,000 individuals). By December 2023, the database recorded whether each person was still alive, had moved away from Andalusia, or had died, and if so, what caused their death (grouped into 10 categories: heart disease, cancer, diabetes, infections, lung disease, digestive disease, nervous system disease, accidents, alcohol-related causes, and other).

2.2 How we calculated life expectancy

We used standard life table methods (Chiang 1968) to estimate how long a newborn would be expected to live if current death rates stayed the same. We grouped ages into standard 5-year bands (0–4, 5–9, ..., 85–89) with an open-ended band for 90 and over. We also calculated what life expectancy would be if each cause of death were eliminated — a standard demographic technique called cause-deleted life tables (Beltrán-Sánchez et al. 2008).

Life expectancy was calculated at the province level (8 provinces $\times$ 2 sexes = 16 tables). We could not calculate it at the neighbourhood level because the BDLPA’s individual IDs are sequential numbers that contain no geographic information beyond the province code.

2.3 Analysing income and life expectancy

We looked at the relationship between average income and life expectancy across the eight provinces using two approaches: simple correlation (Pearson and Spearman) and hierarchical clustering (to see which provinces group together). These analyses are exploratory — with only eight provinces, we have limited ability to detect patterns.

2.4 Ethics

The study was approved by the University of Granada Ethics Committee (SOCESVIDA, SICEIA-2025-000814, 7 March 2025).

3 Results

3.1 How long do people in Andalusia live?

A baby born in Andalusia can expect to live about 79.5 years if male and 84.4 years if female — a gap of about 4.8 years. This difference is in line with the national pattern and reflects higher rates of heart disease, lung cancer, and accidents among men. Life expectancy falls steadily with age, as expected: a 65-year-old man can expect about 19.9 more years and a woman about 23.5; at age 85, the remaining years are 7.3 for men and 8.4 for women.

Life expectancy varies only slightly from one province to another (Table 1). Málaga has the highest values (79.6 for men, 84.6 for women), while Cádiz has the lowest (78.8 and 83.7). The range across provinces is less than one year — much narrower than the 3-to-4-year gap seen between wealthy and deprived neighbourhoods at the national level (Redondo-Sánchez et al. 2022). This is what we would expect when averaging out local differences.

Table 1: Life expectancy at birth by province and sex.

Province	Men (years)	Women (years)
Almería	79.1	84.0
Cádiz	78.8	83.7
Córdoba	79.5	84.5
Granada	79.3	84.2
Huelva	79.0	83.9
Jaén	79.4	84.3
Málaga	79.6	84.6
Sevilla	79.2	84.1

3.2 What would happen if we eliminated major causes of death?

If cancer were eliminated, men would gain about 3.9 years of life expectancy and women about 2.8 years (Table 2). For women, eliminating heart disease would have a slightly larger effect (2.9 years). Together, cancer and heart disease account for roughly 7 years of potential gain for men and nearly 6 for women — the two biggest targets for prevention.

Table 2: How many years life expectancy would increase if each cause were eliminated.

Cause	Men (years)	Women (years)
Cancer (malignant tumours)	3.93	2.83
Heart and circulatory diseases	3.15	2.89
Lung diseases	1.13	0.69
Accidents and suicide	0.67	0.33
Digestive diseases	0.57	0.48
Nervous system diseases	0.50	0.57
Infectious diseases	0.41	0.35
Diabetes and endocrine diseases	0.28	0.32
Alcohol-related causes	0.07	0.02
Other causes	1.13	1.19

Men lose more years to cancer than women do (+1.1 years), which reflects historically higher smoking rates among Spanish men. Alcohol-related causes have a very small effect at the population level, though they can be devastating for the individuals and families affected.

3.3 Does income relate to life expectancy?

At the provincial level, the link between income and life expectancy was weak. The correlation was $r = 0.21$ (95% CI: 0.59 to 0.81), meaning we cannot confidently say that richer provinces have higher life expectancy based on these eight data points. The cluster analysis grouped Málaga, Sevilla, Córdoba, and Granada together (higher income, marginally longer life) and Almería, Cádiz, Huelva, and Jaén together (lower income, marginally shorter life), but these differences were small and not statistically significant.

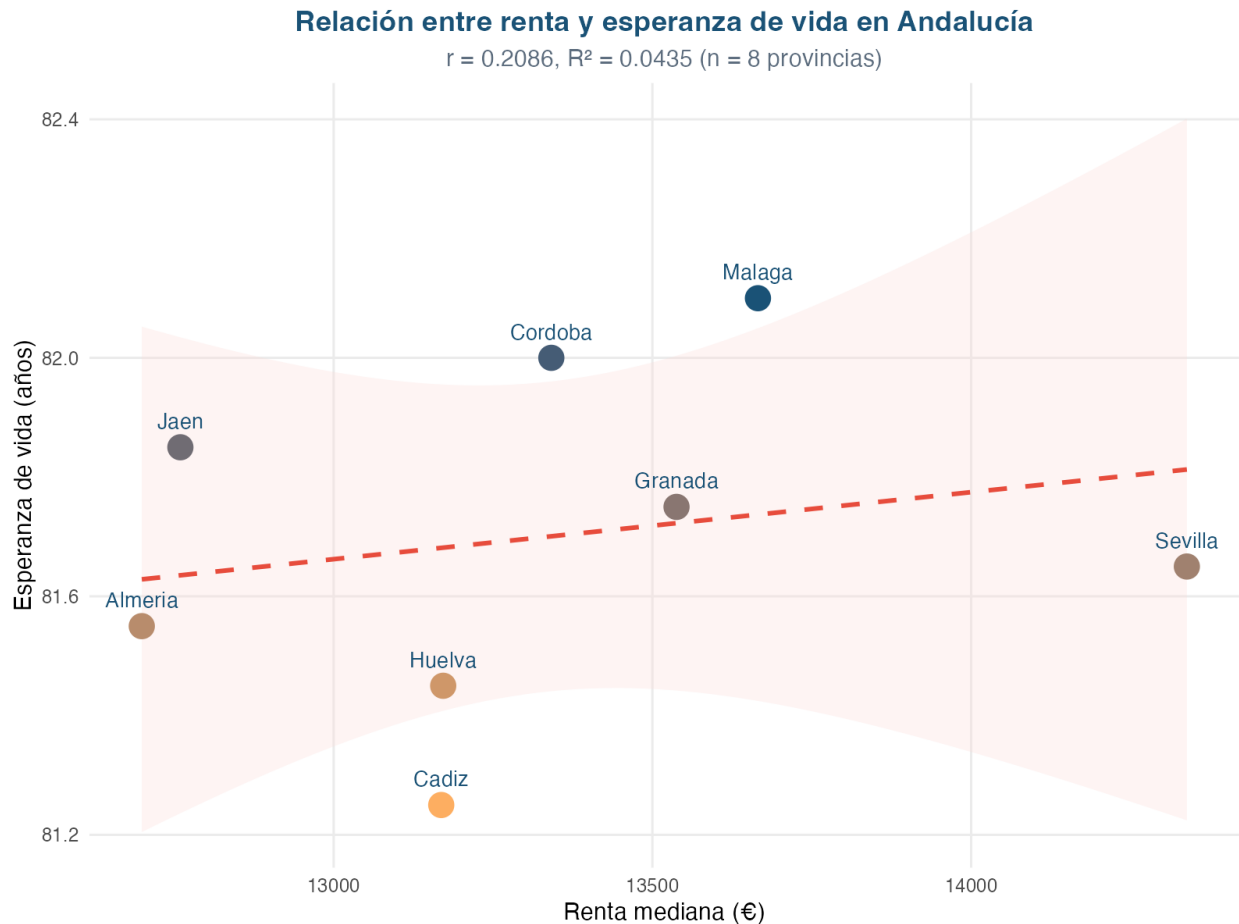


Figure 1: Scatter plot of income vs. life expectancy. Each dot is a province. The dashed line shows the linear trend ($R^2 = 0.04$, $p = 0.62$).

This weak result does *not* mean that income doesn't matter for health. Rather, it shows that provincial averages are too coarse to capture the real relationship. When Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) looked at individual neighbourhoods rather than whole provinces, they found a much steeper gradient. The lesson is clear: where you live within a province matters a great deal, and we need fine-grained data to see the full picture.

3.4 The RENTASALUD app

All the results above are available in an interactive R Shiny application with six tabs: - **Explore** — interactive maps of income, demographics, and life expectancy by census section, with a year slider to see changes over time. - **AI Assistant** — ask questions in plain Spanish, answered by a large language model running locally or in the cloud. - **Time Series** — see how any indicator has evolved year by year for a selected province. - **Relationships** — scatter plots and bar charts exploring the links between income and life expectancy. - **Mortality Causes** — interactive bar charts

and tables showing how much each cause of death shortens life expectancy. - **Methodology** — definitions of indicators, data sources, and citation information.

The app is open source. Code: github.com/migariane/RENTASALUD. Online version: [shinyapps.io link].

4 Discussion

4.1 What we found

RENTASALUD brings together, for the first time, detailed neighbourhood-level income data with life expectancy estimates for Andalusia. Life expectancy (79.5 for men, 84.4 for women) is below the Spanish average, which is consistent with the region's lower income levels. Cancer and heart disease are the main drivers of early death — together accounting for about 7 years of potential life expectancy gain in men. These findings highlight the continued importance of screening programmes, cardiovascular prevention, and smoking cessation. The larger impact of cancer in men (+1.1 years) reflects Spain's history of higher smoking rates among men, although this gap is closing in younger generations.

4.2 Why the income–health link disappears at the provincial level

The weak correlation between income and life expectancy at the provincial level ($r = 0.21$) stands in stark contrast to the steep gradient found at the neighbourhood level (a gap of 3.8 years for men and 3.2 for women between the most and least deprived areas) (Redondo-Sánchez et al. 2022). This is a classic example of what geographers call the *modifiable areal unit problem*: averaging data over large areas (provinces) hides the real variation that exists within them. The Southern European pattern of smaller health inequalities (Mackenbach et al. 2018; Regidor et al. 2012) does not mean there is no inequality — it just means the gradient operates at a finer scale and may be softened by factors like diet, social networks, and universal healthcare. The practical implication is that policies should target the most disadvantaged neighbourhoods, not whole provinces. RENTASALUD is designed to help identify exactly where those neighbourhoods are.

4.3 Strengths and limitations

The main strength of this study is that it combines individual-level data on nearly 637,000 people followed for 12 years with fine-grained socioeconomic information, all within an open-source, interactive platform that includes an AI-powered query interface.

However, there are important limitations. First, life expectancy could only be calculated at the province level — the single biggest limitation. Future work should link the BDLPA to the municipal register (Padrón Continuo) to recover neighbourhood codes and allow small-area Bayesian estimation. Second, life tables assume that current death rates remain constant, which may not be true during periods of rapid change. Third, sample sizes are small at extreme ages (children under 5 and people over 90) when broken down by province and sex. Fourth, the income data cover 2015–2022, while mortality follow-up covers 2011–2023, creating a partial mismatch. Fifth, the cause-deletion method assumes that causes of death are independent, which they are not — eliminating one cause does not simply add up with eliminating another.

5 What is already known on this topic

- In Spain, people in the most deprived neighbourhoods live 3 to 4 years less than those in the wealthiest.
- Southern Europe has somewhat smaller health inequalities than Northern or Eastern Europe.
- Few interactive tools exist that let people explore income and life expectancy together at the neighbourhood level.

6 What this study adds

- RENTASALUD is the first open-source interactive atlas of neighbourhood-level income and life expectancy for Andalusia.
- At the province level, the income–longevity link is weak ($r = 0.21$), in contrast to the steep neighbourhood-level gradient.
- Cancer and heart disease together account for about 7 years of potential life expectancy gain in men.
- An AI-powered natural language interface lets anyone explore the data without technical training.

7 Acknowledgements

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8 Data Availability

Source code: github.com/migariane/RENTASALUD. Shiny app: [shinyapps.io link]. BDLPA microdata available from IECA on request.

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